

Snowmaking and Day-Ahead Load Forecasts

A Pre-Registered Test in Four Markets

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Independent research
August 2026
[Code and data on GitHub](#)

Abstract

The question. Austrian ski resorts burn about **281 GWh** a season making artificial snow, almost all of it on cold November and December nights: **8-15% of overnight demand**. The guns run only below a wet-bulb temperature near $-2\text{ }^{\circ}\text{C}$ and stop once the base layer is built, so how much power a cold night draws depends on how much snow has already been made.

The answer. No. The prediction and the stopping rules were committed **before any load data was opened**. On thirteen seasons and 780 nights the effect is **+5.1 MW, give or take 11.9**, and two of three rules fired. The same model on the same nights finds the Christmas industrial shutdown at **-274 MW**, so it can see effects of the size snowmaking would have to produce. The load is **absorbed** by the forecast rather than missed by it.

Elsewhere, and in the market. Italy-North agrees at **+4.3 $\pm$ 10.2**. Switzerland could never have found the effect. Vermont finds what was predicted but sits on its own detection threshold and does not survive a change of weather station. A pre-registered price test cannot tell either: overnight the supply stack is so flat that Austria's entire fleet is worth about **€14/MWh** against a spread whose swings are three times that.

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1. The question

Transmission system operators publish a day-ahead load forecast, and both it and the realised load are free. If snowmaking is invisible to the model, that error should be positive on nights when snowmaking runs, and the size of the miss should depend on the snowpack rather than on temperature alone. The interest is whether a large, lumpy, path-dependent industrial load can hide inside a production forecast, a structure that recurs in Spanish irrigation pumping and North American grain drying.

2. How big is the load

From a 2026 survey of 141 Austrian resorts (30 usable, 4,253 equipped hectares, 34.0% of Austrian ski volume), extrapolated nationally:

Table 1. Published snowmaking figures used throughout.

Quantity	Value
Season electricity, Austria-wide	281 GWh (260–309)
Share of Austrian electricity consumption	0.46%
Mean operating hours per snowmaker per season	184.6 h
Snowmakers per hectare	2.9
Energy per hectare equipped	22,449 kWh
Energy per m ³ of snow	3.3 kWh

Instantaneous power follows from energy over operating hours. The coincident ceiling is $281 \text{ GWh} \div 184.6 \text{ h} = \mathbf{1.52 \text{ GW}}$, a mean 41.9 kW per snowmaker while running. At 40–70% coincidence the national draw is 0.61–1.07 GW against a November–December overnight load of 7.0–7.5 GW, so snowmaking is **8–15% of overnight demand**.

3. Why a forecast might miss it, and why it might not

The naive version is that load forecasts are temperature models and temperature models have no memory. Two mechanisms argue against it, both written down before the data was opened. **A memoryless model still absorbs the average response:** the temperature coefficient is fitted on history in which cold nights are snowmaking nights, so the residual holds the snowmaking anomaly rather than the load. **Production forecasts are autoregressive:** APG publishes at 08:00 on historical actual load, day type and temperature, so a lagged-load term carries a running campaign into tomorrow. Both push α , the share left unexplained, down, and α binds the whole design.

4. Identification

The obvious test, comparing nights just below the wet-bulb threshold to nights just above, is contaminated. At 1,800 m, $-2 \text{ }^{\circ}\text{C}$ wet bulb is roughly -1 to $0 \text{ }^{\circ}\text{C}$ air temperature, and every nonlinearity in heating load lives near freezing: heat-pump COP collapse, defrost cycles, trace heating. Humidity, which enters wet bulb, is itself a load driver.

The identifying variation is therefore not the threshold crossing but its interaction with accumulated season-to-date cold:

```

err ~ below × cum_cold_hours
      + dist + below:dist + holiday
      + doy + doy2 + season FE + dow FE

err          = actual load - day-ahead forecast, MW
below        = 1 if alpine wet-bulb index < -2 °C
dist         = wb_index - (-2)
cum_cold_hours = hours below threshold since 1 Oct, current season

```

Heating load does not care how many cold hours the season has already delivered. Snowmaking does, because the base gets built and the guns stop. The interaction is the only coefficient in this design a temperature confound cannot produce.

4.1 Building the wet-bulb index

Three choices decide the answer before the econometrics do. Stations are selected by **altitude band** (900–2,600 m), because valley stations cross the threshold hundreds of hours later than the places snow is made. Wet bulb is corrected for **station pressure**, since ISA pressure at 1,800 m is 815 hPa and assuming sea level shifts it by 0.2–0.4 °C. And the psychrometric equation is **solved** rather than approximated, because the Stull (2011) closed form errs by 0.7–1.0 °C below freezing.

5. Pre-registration

Primary prediction. The `below × cum_cold_hours` coefficient is **negative** and significant at 5%.

Negative, not positive, and the direction is the whole test. Snowmaking adds load, so a cold night on its own should push the forecast error up. What identifies snowmaking rather than heating is that the push *fades*: by late December the base is built and the guns are off, so the same weather draws far less power than in November. A fade is a negative number, and a positive interaction is the wrong shape rather than a bigger effect.

Kill criteria, committed before looking:

- The interaction is zero or positive with a tight confidence interval → no memory effect. Stop. · *fired*
- The event-study profile is flat across campaign days → the forecast already absorbs it. Stop. · *fired*
- A cold, flat placebo country shows the same jump → the result is heating load. Stop. · *not run, moot once the first two fired*

All outcomes were committed to publication in advance. A null was named as the modal outcome.

6. Statistical power

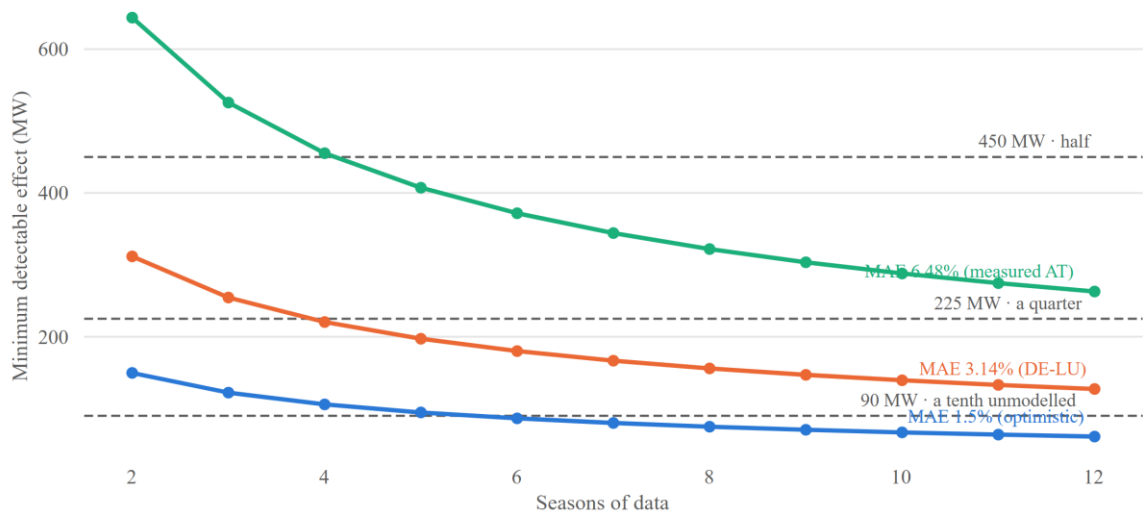


Figure 1. Minimum detectable effect against seasons of data, for three assumptions about day-ahead forecast error, with the plausible residual signal overlaid. The pass mark is how much of the ~900 MW snowmaking load the forecast must leave unexplained. Additional seasons barely move it; α is the binding constraint.

Measured on the data, the Austrian forecast is noisier than the German-Luxembourg benchmark of 3.14% used to calibrate this: **6.48% MAE** on November–December night hours, 554 MW after fixed effects, raising the pass mark to $\alpha \geq 27.4\%$. Section 8.1 redoes it on fitted standard errors, which puts the real Austrian bar at 47%.

7. Results

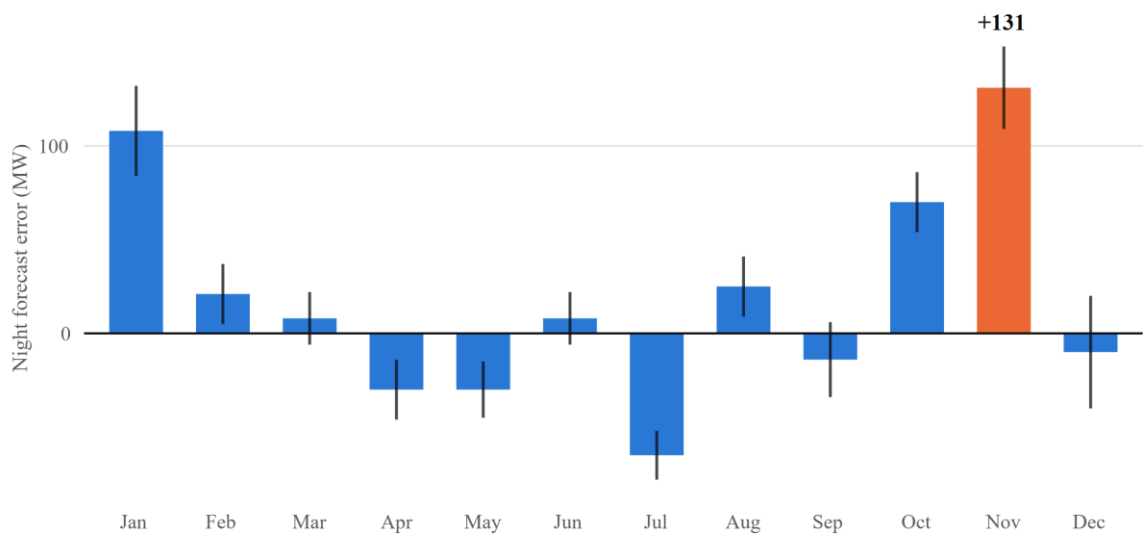


Figure 2. Night forecast error by month, 2010–2022, night-level observations. November is the most under-forecast month of the year at $+131 \pm 22$ MW. Bars are ± 1 s.e.

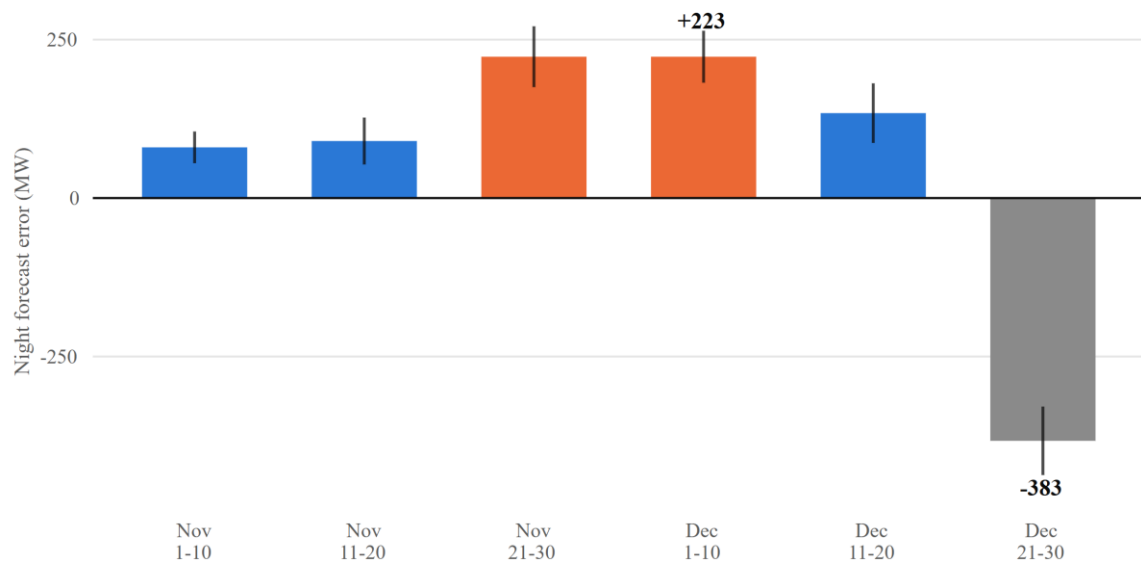


Figure 3. The same error in 10-day bins across November and December. The bias climbs to $+223 \pm 41$ MW in early December and then collapses at the Christmas industrial shutdown. This is the shape, magnitude and timing snowmaking would produce, and also what a seasonal heating ramp produces.

The unconditional seasonal profile was encouraging. November is the most under-forecast month of the year and the bias climbs to $+223 \pm 41$ MW in early December before collapsing at Christmas: the right shape and timing for snowmaking, and also what a seasonal heating ramp produces. Conditional on weather, it vanishes.

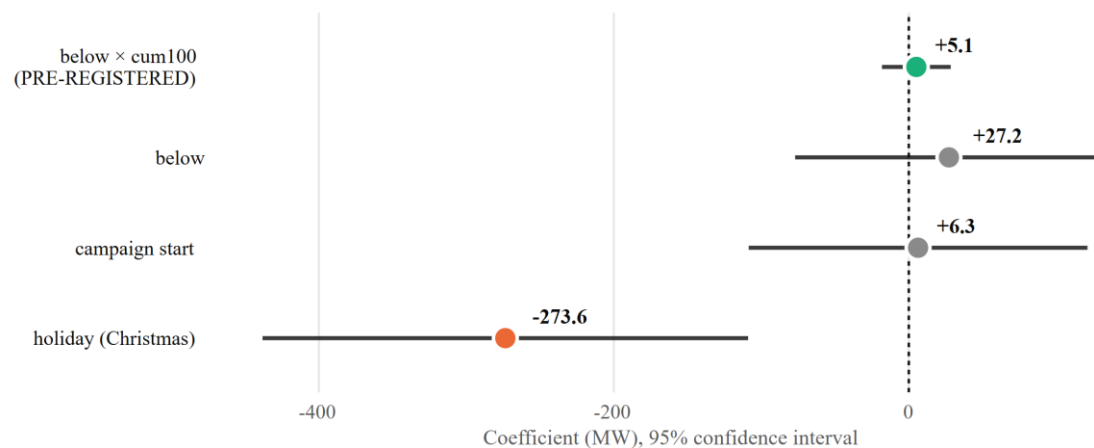


Figure 4. Coefficients from the primary specification with 95% intervals. The pre-registered interaction sits on zero. The same model, on the same nights, places the Christmas shutdown at -274 MW, so the design is not blind to effects of the size snowmaking would have to produce.

Table 2. Coefficients in MW, HC1 standard errors in parentheses, night-level observations. The primary coefficient is zero in every specification. The holiday control is strongly significant in every specification.

Specification	n	below × cum100	below	holiday
Nov–Dec, all nights	780	+5.1 (11.9)	+27.2 (53.2)	–273.6 (84.0)
Bandwidth wb+2 ≤ 3 °C	418	+4.2 (14.7)	+39.8 (83.1)	–210.0 (121.2)
With campaign-start dummies	780	+5.4 (12.0)	+23.1 (55.1)	–273.7 (84.0)
Seasons 2016–2022 only	420	–7.9 (11.5)	–3.3 (49.5)	–372.0 (92.7)

The estimated equation is not the registered one, so both were run. The registered version uses hourly rows with hour fixed effects and date-clustered errors, a 21:00–05:59 night, a ± 3 °C bandwidth, and no holiday or day-of-season controls. Run unchanged it gives **–1.3 (12.3), p = 0.91** against the **+5.1 (11.9)** above: the same null at the same precision, carrying the predicted negative sign at a ninth of its own standard error.

Table 3. The scenario where α should be highest, because autoregressive terms have not yet caught up. Both dummies enter the same specification. The profile is flat, which is what the kill criterion names. The first night's +6.3 is in the predicted direction and is a ninth of its own standard error; the second night is wrong-signed and equally insignificant.

Campaign-start terms	Coef.	s.e.	t
First night of a cold snap	+6.3	58.6	0.11
Second night	–91.7	86.7	–1.06

7.1 Why this is a null rather than an absence of evidence

The specification is not underpowered for effects of the relevant size. On the same 780 nights it recovers the Christmas shutdown at –274 MW with $t = -3.3$, and the snowmaking interaction is $+5 \pm 12$. Two limits: the Christmas argument works because APG leaves the shutdown in its residual, which Terna and Swissgrid do not, and it is a claim about 274 MW. The smallest seasonal swing the fitted model could have detected at 80% power is 426 MW against a fleet drawing an estimated 900 MW, so Austria could only ever have caught a forecaster missing 47% or more.

8. Was Austria a badly chosen case?

A null is only interesting if the test was fair. Ranking systems by snowmaking energy per gigawatt of winter overnight load puts Austria near the top of the world. The table is desk scoping, published or derived snowmaking energy over published system load, and three of its rows have since been tested for real in section 8.1.

Table 4. Worldwide ranking. *Derived by scaling equipped hectares at the Austrian intensity of 22,449 kWh/ha; no published national total exists outside Austria, Switzerland and France. †Germany's figure covers lifts and snowmaking together.

System	GWh/season	O/n load (GW)	Ratio	Free forecast?
ISO-NE Vermont region (<i>tested</i>)	40–90*	0.59	68–153	Yes, per region
Austria (this study)	281	6.6	43	Yes
Italy-North (<i>tested</i>)	~560*	16.2	~35	Yes, Terna
ISO-NE New Hampshire	22–54*	~1.25	18–43	Yes
PSCO (Colorado)	45–70*	~4	11–18	Yes, EIA-930
Switzerland	60–65	~8.4	7.1–7.7	Yes, Swissgrid
France	>110	~60	1.8	Yes, ODRÉ
Germany	≤43†	~52	≤0.8	Yes, SMARD

8.1 Three of those rows were then tested

Same specification, same wet-bulb solver with the station-pressure correction, same 20:00–06:59 night, same fixed effects. Only the load and weather sources change.

Table 5. Replications. Coefficients in MW, HC1 standard errors in parentheses. Vermont's outcome is the regional *share*; its native coefficient is –0.0211 percentage points per 100 cold hours, converted here at the 120 MW one point of share is worth.

System	Seasons	Nights	below × cum100	Swing found	Detectable	Needs α	Verdict
Italy-North (Terna)	7	420	+4.3 (10.2)	+61	403	27%*	Null, and uncertified
Vermont (ISO-NE)	5	297	–2.5 (0.9)	–25	24	34%	Predicted sign, on its own threshold
Austria	13	780	+5.1 (11.9)	+65	426	47%	Null
Switzerland	9	540	–42.5 (14.3)	–427	402	201%	Not interpretable

How to read the power columns. The coefficient is per 100 accumulated cold hours and a season delivers about a thousand, so the seasonal swing is what has to be detectable, and "needs α" is that divided by each market's coincident snowmaking draw. Austria's 900 MW comes from a published survey and Italy's 1,500 MW is a desk derivation, so a five-point gap there is not a real difference.

No test here could see a forecaster missing less than about a quarter of the snowmaking load, and Austria, on the best-grounded fleet estimate of the four, could not see one missing less than half. That is the finding behind all the individual findings.

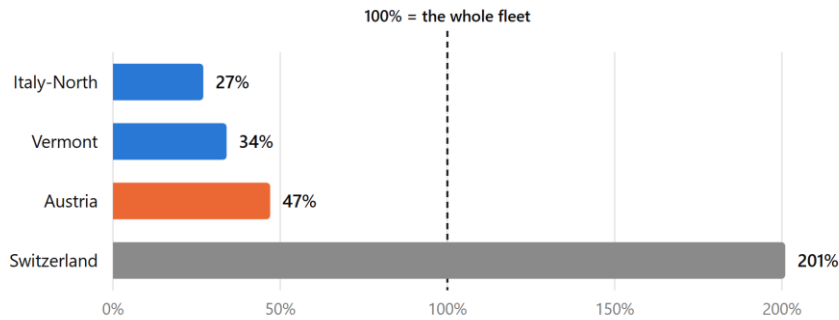


Figure 5. The detection threshold, market by market: how much of its own snowmaking fleet a forecaster has to be missing before this design can see anything. Switzerland sits past the 100% ceiling, so a Swiss forecaster modelling none of Swiss snowmaking would still be invisible to the test. Austria, on the only fleet figure from a published survey, needs 47%.

Italy-North replicates the null at +4.3 (10.2) against Austria's +5.1 (11.9). **The Christmas sanity gate is an Austrian regularity**, working only because APG's night MAE is 6.48%; Terna's is 2.17% and its forecast predicts -4,461 MW of a -4,482 MW shutdown, so the Italian null carries only a paper-power bound. **Switzerland could never have detected the effect** at a required α of 201%, and its significant, correctly signed coefficient implies a swing larger than the entire Swiss fleet, so it is a confound.

Vermont meets the prediction, then fails to hold it. ISO-NE's regional forecast is not a fixed load-share allocation, so the share is what the test targets. On five seasons the interaction is -0.0211 percentage points of share per 100 cold hours (HC1 0.0073, $p = 0.004$), with Rhode Island, a same-forecaster zero-snowmaking placebo in the same feed, null at +0.0019 (0.0037). But Mount Washington in place of the road stations gives -0.0058 (0.0068), the effect sits on its noise floor at -25 MW against 24 MW detectable, and ISO-NE's own zonal report drops it to -0.0281 (0.0188) while firing the placebo. Vermont is suggestive rather than confirmation.

8.2 The price test: overnight, those megawatts are worth almost nothing

A load forecast error is the grid operator's error, not the market's, so a null on load does not rule out a price effect. That test is pre-registered and run in `src/price/`. It is on the **day-ahead spot auction**, the only instrument that clears on a snowmaking decision's eighteen-hour horizon. A quarterly baseload futures contract averages away the day-to-day wet-bulb variation the design runs on, so no futures contract is tested. The outcome is the night-minus-midday price spread, which differences out the fuel cost that dominates the level through the crisis winters; the right-hand side is the load test's, unchanged.

Table 6. The price power gate, which prints before any coefficient by design. Supply slope in €/MWh per GW, estimated from each market's own night residual load. Everything else in €/MWh.

Market	Seasons	Nights	Supply slope	Fleet	Worth	Detectable	Short by
Austria	5	300	+15.6	900 MW	+14.0	18.0	1.3×
Italy-North	7	420	+3.5	1,500 MW	+5.3	14.4	2.7×
Switzerland	9	540	+7.7	200 MW	+1.55	5.9	3.8×
Vermont (ISO-NE, USD)	6	344	+23.9	70 MW	+1.67	6.7	4.0×

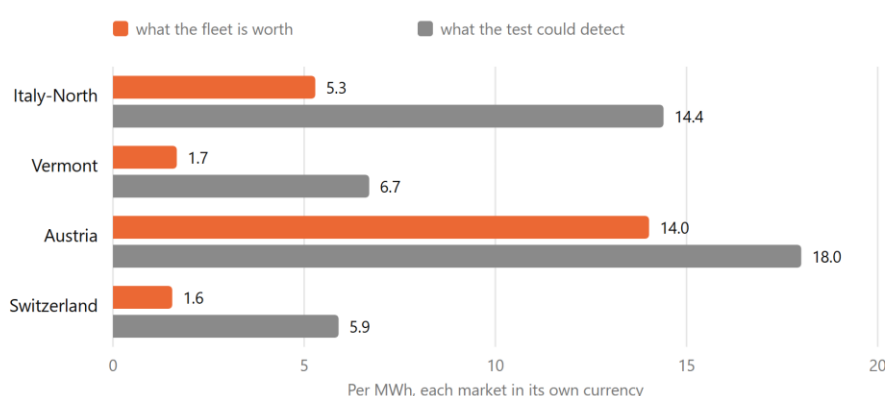


Figure 6. Why the price test cannot answer the question. In every market the grey bar is longer than the orange one: the smallest price effect the test could have detected is larger than what the entire snowmaking fleet is worth at the overnight margin. Austria comes closest, short by 1.3×, and gains a season of price data every winter.

All four come back underpowered. The overnight merit order is nearly flat, so Austria's own night data prices a 900 MW fleet at about €14/MWh at the margin against a night-day spread with a €43/MWh standard deviation. A load test can find a few hundred megawatts because load is measured in megawatts; a price test has to find what they are worth. Austria is the near miss at 1.3×. The Christmas gate fails here for a second, independent reason: Christmas removes midday and night demand at once, so an outcome built to ignore common demand shocks ignores the reference effect too.

In Vermont a placebo settled it. Vermont's price interaction is **−1.47 (0.54), p = 0.007**, the predicted sign, and Rhode Island, which makes no snow, returns **−1.33 (0.52), p = 0.011**. That is a New England-wide relationship between accumulated cold and the spread, not a Vermont signal. It bites where the load placebo could not, because the eight regional shares sum to zero by construction and force a genuine Vermont effect to push Rhode Island the other way.

8.3 A different treatment: snowfall-forecast revisions

Every test above runs on accumulated cold hours, a slow seasonal variable badly collinear with day-of-season, and that collinearity is what produced required α between 27% and 201%. A fourth arm replaces it with **NWS snowfall-forecast revisions**: high-frequency news arriving six to ten times a day rather than nine times a season, and a revision to expected *natural* snowfall holding temperature fixed should not move heating demand at all. Because ISO-NE's day-ahead

market closes mid-morning, a revision arriving after the close is information the price cannot contain, which makes a real falsification test available.

On 600 nights across ten Vermont seasons that falsification passes: news postdating the auction moves nothing, **-0.86 (2.58)**. The pre-gate coefficient is **+0.20 (2.20)**, wrong-signed and underpowered by six to eleven times, so it is reported and not read. What the arm adds is a treatment with real high-frequency variation and a falsification test that passes. Full write-up in `src/revision/README.md`.

9. Limitations

- **The published forecast is the operator's transparency artefact, not the trading consensus.** A bias in it shows a blind spot in APG's forecast, **not** a market mispricing.
- **The treatment is a switch and the physics is a staircase.** Olefs et al. (2010), the source of the $-2\text{ }^{\circ}\text{C}$ threshold, defines a *relationship between wet-bulb temperature and snowmaking capacity* alongside it. A binary treatment discards the intensity margin, and much of the megawattage lives there.
- **Water temperature mimics the predicted effect.** Production at the margin needs water below about $+2\text{ }^{\circ}\text{C}$ and reservoir water cools through the season, so an identical wet bulb is more productive in late December. That needs no path dependence, and nothing here rules it out.
- **No working sensitivity check exists outside Austria**, so every replication after the first carries a paper power calculation and no proof its instrument can see anything. `cum_cold_hours` also ignores melt and water remaining, operators skip marginal nights just below threshold, APG's load excludes a Vorarlberg corridor, the opening-date test was not identifiable from calendar dates ($R^2 = 0.798$), and the NL/DK placebo was not run.

10. Data and reproduction

No API token is required. Austrian Power Grid publishes both load series back to 2009 as nested per-year ZIP archives with no registration, and GeoSphere Austria's station API needs no key. One script reproduces everything:

```
git clone https://github.com/baptistecristo/Electricity-demand-forecasting-in-Austria
pip install -r requirements.txt
python src/apg_pipeline.py
```

For cross-country work, `energy-charts.info` serves day-ahead forecast, load and spot price for most European bidding zones with no token. Futures settlement history is paywalled at EEX and ICE, so day-ahead spot is the honest substitute.

References

- [1] Aigner, Steiger & Mayer (2026). Snowmaking in Austria: resource consumption and greenhouse gas emissions. *Journal of Sustainable Tourism*. [figures](#)
- [2] Olefs, Fischer & Lang (2010). Boundary conditions for artificial snow production in the Austrian Alps. *J. Appl. Meteorol. Climatol.* 49(6). [link](#)

- [3] Stull (2011). Wet-bulb temperature from relative humidity and air temperature. *J. Appl. Meteorol. Climatol.* 50(11). Used as a benchmark, not in the pipeline.
- [4] Austrian Power Grid. [Actual total load](#) and [total load forecast](#).
- [5] GeoSphere Austria. [Dataset API](#).
- [6] Maldonado et al. [arXiv:2302.11017](#) , DE-LU day-ahead load forecast MAE.
- [7] ISO New England. [Three-day reliability region demand forecast](#).
- [8] Fraunhofer ISE. [energy-charts API](#).

Pre-registration and results are separate commits in the repository; the commit history establishes that the stopping rules predate the data. Code and figures are MIT-licensed. github.com/baptistecristo/Electricity-demand-forecasting-in-Austria